

JAGDAMBA PROFILE ERP

Ultimate Technical Specification

A deep-dive into the architectural integrity, modular structure, and technological stack of the enterprise ecosystem.

Generated: May 4, 2026 | Technical Authority Edition

1. Comprehensive Tech Stack

The application is engineered using a modern, multi-layered architecture designed for high performance and strict data integrity.

Frontend Core

- React 18.3: Component-based UI with Concurrent Rendering.
- TypeScript 5.x: Strict type definitions for enterprise-grade reliability.
- Vite 6.0: Lightning-fast build pipeline and Hot Module Replacement.

Data & Persistence

- Better-SQLite3: Native SQLite driver for high-speed local data handling.
- Prisma ORM: Type-safe database client and schema management.
- Context API: Unidirectional data flow for global state orchestration.

UI/UX System

- Tailwind CSS 4: Utility-first styling with high-impact design tokens.
- Lucide React: Vector-based iconography for visual clarity.
- Framer Motion: Advanced micro-interactions and transitions.

2. Module Encyclopedia

A. Order Management & CRM

Manages the entire sales lifecycle from initial lead to Purchase Order generation.

- CRM: Industry-specific client database with interaction logging.
- Order Logic: Dynamic line-item calculation with real-time tax/total updates.

B. Production ERP

The core workflow engine tracking orders through 11 physical manufacturing stages.

- Stages 1-3: Nesting & Planning (Optimization logic).
- Stages 4-7: Laser & Plasma Processing (Production tracking).
- Stages 8-11: Dispatch, Logistics, and Final Reconciliation.

C. HR & Logistics

Enterprise Resource Planning for human and physical capital.

- Staffing: Departmental silos (Office, Production, Accounts).
- Fleet: Vehicle capacity tracking and transporter performance logs.

3. Data Models & Interfaces

The system architecture relies on standardized TypeScript interfaces.

Order Entity Schema

```
interface Order {  
  id: string;  
  orderNo: string;  
  partyName: string;  
  items: OrderItem[];  
  status: 'PENDING' | 'PRODUCTION' | 'DISPATCHED';  
  handledBy: string;  
}
```

Staff Entity Schema

```
interface Employee {  
  id: string;  
  name: string;  
  department: 'OFFICE' | 'PRODUCTION' | 'ACCOUNTS';  
  skills: string[];  
  status: 'ACTIVE' | 'ON_LEAVE';  
}
```

